Product Requirements Document: StudyBearer

Hackathon Track: iSITE AI Hackathon 2026 - AI in Education (Theme 1: Learning... Without 'Learning')

App Name: StudyBearer

Core Philosophy: Automating the friction of studying so students can focus on the learning.

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1. Executive Summary

Students are frequently handed massive PDF modules and left to their own devices, leading to poor time management, overwhelming anxiety, and a high risk of falling behind in complex subjects. **StudyBearer** transforms passive documents into active, gamified learning journeys. By uploading a PDF and defining their availability, students receive an automated, AI-chunked study roadmap integrated directly into their Google Calendar, followed by gamified quizzes.

StudyBearer makes education feel effortless by completely automating the "planning" and "testing" phases, leaving only the "discovering" phase for the user.

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2. Problem Statement & Empathy (User-Centric Design)

Traditional studying demands high executive function: breaking down chapters, scheduling study blocks, and creating practice tests.

- **The Pain:** Students dealing with punctuality issues, poor attendance records, or the weight of previously failed classes (like heavy technical subjects such as Data Structures) often struggle not because they lack capability, but because they lack a structured, automated roadmap. A 100-page PDF is intimidating; a 30-minute scheduled block on "Looping Constructs" is approachable.
- **The Need:** A system that empathetically holds the student's hand, integrating seamlessly into the tools they already use daily (Google Calendar) rather than forcing them to adopt a completely new ecosystem.

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3. Key Features & User Flow

3.1. Onboarding & Auth

- Simple, frictionless authentication using Google OAuth to tie directly into the user's calendar ecosystem.

3.2. The Intake (Upload & Parameters)

- User uploads course materials (PDF format).
- User inputs their available study hours and a target completion date.

3.3. The AI Orchestration (n8n Magic)

- The system parses the document, extracting key learning objectives.
- It segments the material into logical, bite-sized daily modules (e.g., Monday: Linked Lists, Wednesday: Binary Trees).

3.4. Calendar Sync

- The segmented roadmap is automatically pushed to the user's Google Calendar as scheduled events, removing the friction of manual time management.

3.5. Gamified Recall (The Quizzes)

- At the end of a scheduled study block, the app unlocks a highly engaging, AI-generated quiz specific to that day's topic.
- Formats include Multiple Choice, Enumeration, and Identification.

3.6. Progression Engine (Streaks & Achievements)

- **Consistency Combos (Streaks):** Completing the daily scheduled study block and passing the quiz increases the user's "Combo." Missing a scheduled day breaks the combo, creating strong loss-aversion.
- **Ranked Milestones & Achievements:** Users climb ranks and unlock premium profile banners.
  - *First Blood:* Complete the very first AI-generated study block.
  - *Flawless Victory:* Score 100% on a post-study quiz.
  - *Competitive Ladder:* Users accumulate Elo points based on quiz accuracy and streak multipliers, climbing from base tiers up to elite statuses like Mythic, or grinding to hit a 2000+ Elo rating. This turns a semester-long subject into a competitive grind.

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4. Design & Aesthetics

To distance the app from the sterile, boring look of traditional academic portals, StudyBearer utilizes a **dark-toned, cinematic, and moody visual aesthetic**.

- **The Vibe:** This premium, sleek interface feels more akin to a modern developer tool or a high-end gaming interface, reducing the cognitive dread associated with "school software."
- **Unlock Animations:** Earning an achievement features sleek, high-contrast animations (e.g., subtle neon glows against a dark background).
- **Customization:** Unlocked achievements grant the user equipable titles and dark-mode avatar borders, making their dashboard feel like a customized player card.

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5. Technical Architecture

The application is built for extreme performance on the frontend, relying heavily on automated workflow orchestration for the heavy lifting.

5.1. Web Application Stack

- **Framework:** Astro (Islands architecture allows for a blazing-fast static shell with interactive React components only where necessary, like the quiz interface).
- **UI Library:** React (for complex state management in the dashboard and quiz engine).
- **Backend & Auth:** Supabase (PostgreSQL database, secure file storage for PDFs, and instant Google Auth).
- **ORM:** Prisma (for rapid, type-safe database queries, ensuring high velocity during the hackathon).
- **Database Schema:** Includes a UserStats model to track current_streak, longest_streak, total_points, and a relational UserAchievements table.

5.2. The n8n AI Engine (Core Focus - 20% Rubric)

Because the usage of n8n is a heavily graded criterion, the web app functions essentially as a beautiful terminal. **n8n acts as the absolute central nervous system** of StudyBearer.

n8n Workflow	Trigger	Execution Steps (Nodes)
PDF Processing & Roadmap	Supabase Webhook (on file upload)	1. **HTTP Node**: Fetch PDF from Supabase. 2. **Document Parser**: Extract text. 3. **AI Agent (Gemini/OpenAI)**: Chunks text into a JSON roadmap. 4. **Postgres Node**: Save roadmap to Supabase.
Calendar Orchestration	Triggered post-Roadmap creation	1. **Data Transformer**: Iterates through JSON array. 2. **Google Calendar API Node**: Creates calendar events for each topic.
Quiz Generation	Webhook (User clicks "Start Quiz")	1. **Postgres Node**: Retrieves topic. 2. **AI Agent**: Generates structured JSON quiz (MCQ, Enumeration, ID). 3. **Webhook Response**: Returns JSON to React frontend.
Gamification & Evaluation	Webhook (User submits Quiz)	1. **Postgres Node**: Retrieve streak data.

n8n Workflow	Trigger	Execution Steps (Nodes)
		2. **Logic Node**: Check if submission is within the 24h window (increment or reset streak). 3. **Condition Node**: Check for milestone/perfect score. 4. **Postgres Node**: Update UserStats & UserAchievements. 5. **Webhook Response**: Return unlock_events to frontend for cinematic animations.

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6. Hackathon Rubric Alignment

- **Impact & Relevance (30%):** Directly answers Theme 1 by shifting the burden of "planning" to an AI. Education becomes an act of discovery guided by the app, eliminating the friction of "traditional studying."
- **Feasibility & Execution (20%):** The Astro + Supabase + Prisma stack allows for rapid frontend development. Abstracting the complex API calls (LLMs, Calendar, parsing, logic loops) into n8n drastically reduces backend coding time, ensuring a practical, working prototype.
- **User-Centric Design (10%):** Grounded in the real-world struggles of students facing academic burnout. It validates the psychological need for immediate gratification through gaming mechanics. The Google Calendar integration respects existing habits, and the dark-mode aesthetic reduces eye strain.

- **Presentation & Pitch (20%):** The pitch writes itself: *"We don't just give you a summary of a PDF. We organize your life, schedule your week, and quiz you like a personal tutor, all while you sleep."*
- **Usage of N8N Platform (20%):** n8n is not just an API wrapper; it is the entire backend brain and rules engine. It manages multi-step AI orchestration, 3rd party API integration, and complex streak/achievement logic.